Recent Publications

01.2026

Publication: BlackWall: Protect an AI from going rogue via an AI

Technische Universität Bergakademie Freiberg, Freiberg, Germany

- Proposed an asynchronous firewall architecture for real-time AI traffic analysis.
- Designed custom filtering rules to detect and mitigate rogue agentic behaviors.
- Validated engine efficiency in high-throughput environments against standard benchmarks.
- Available at: https://www.researchgate.net/publication/400669946_BlackWall_-Protect_an_AI_from_going_rogue_via_an_AI

07.2025

Publication: From Words to Feelings – Culture-Aware Emotion AI

Technische Universität Bergakademie Freiberg, Freiberg, Germany

- Developed a multimodal LLM framework for emotion prediction from text, audio, and video.
- Leveraged cross-modal embeddings to interpret culturally specific emotional cues.
- Implemented knowledge distillation for robust performance on degraded data modalities.
- Available at: https://www.researchgate.net/publication/400669787_Moodium_From_Words_to_Feelings_An_LLM-Based_Culturally_Aware_Framework_for_Emotion_Classification_and_Prediction

06.2025

Publication: Financial Forecasting with Scientific Machine Learning

Technische Universität Bergakademie Freiberg, Freiberg, Germany

- Applied SINDy algorithm to model nonlinear dynamics in financial time series.
- Extracted interpretable differential equations from noisy market data.
- Enhanced forecast accuracy by integrating domain knowledge with feature engineering.
- Available at: https://www.researchgate.net/publication/393173017_Financial_Forecasting_Equations_with_Scientific_Machine_Learning

01.2025

CAPTCHA Unmasked: The Math That Outsmarts Bots

Technische Universität Bergakademie Freiberg, Freiberg, Germany

- Optimized CAPTCHA difficulty parameters using gradient descent and weighted cross-entropy loss.
- Extracted hierarchical visual features for classification using Convolutional Neural Networks (CNNs).
- Obscured structural patterns via affine transformations, nonlinear warping, and color segmentation.
- Available at: https://www.researchgate.net/publication/392626868_CAPTCHA_Unmasked_The_Math_That_Outsmarts_Bots

Topic: Design and Implementation of Proactive, Multimodal Cognitive Architecture for Autonomous Personal Agents - AGI

Abstract & Problem Statement

Current LLMs are fundamentally reactive "brains in a jar," limited by episodic amnesia and a lack of environmental integration. This research proposes a Proactive Agentic Framework to transition from passive prompting to autonomous assistance. By integrating an "always-on" multimodal perception loop with a hierarchical memory system, the framework enables agents to self-initiate actions based on environmental cues, such as user fatigue, while resolving the Alignment-Autonomy Paradox through real-time ethical guardrails.

Proposed Methodology & Architecture

The study introduces a modular Cognitive Operating System driven by a continuous four-stage cycle. A perception layer utilizing YOLO-World and Silero VAD filters environmental noise, gating data for a Cognitive Core that utilizes Reflexion and Tree of Thoughts for self-correcting logic. To ensure long-horizon context, the system employs GraphRAG for structured memory retrieval, all governed by a "Constitutional AI" supervisor that vets autonomous plans against ethical constraints before execution.

Technology Stack and Implementation

The implementation leverages LangGraph (with integration with Python) to manage real-time, cyclic workflows. Multimodal inference is handled by a hybrid of vLLM. To validate the superiority of structured memory for personalization, the data layer integrates Neo4j and Weaviate, testing the hypothesis that knowledge graphs provide more reliable long-term agent behavior than standard vector search.

Key References

1. Wang, L., et al. (2024). A Survey on Large Language Model-based Autonomous Agents.
2. MemGPT Team. (2024). MemGPT: Towards LLMs as Operating Systems.
3. Edge, D., et al. (2024). From Local to Global: A Graph RAG Approach to Query-Focused Summarization.
4. Shinn, N., et al. (2023). Reflection: Language Agents with Verbal Reinforcement Learning.
5. Anthropic Research. (2024). Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training.
6. Meta AI. (2024). Llama Guard: LLM-based Input-Output Safeguard Model.

Topic: Context-Aware Augmented Reality: An Agentic Multimodal Framework for Real Time Anomaly Detection in Industrial Environments

Abstract & Problem Statement

Traditional computer vision simply puts labels on objects but misses the bigger picture. In high pressure environments like disaster zones, users often feel swamped with raw data because their tools see a valve but cannot tell if it is leaking or dangerous. To fix this gap, we created the Agentic Multimodal Perception framework. This system takes input from various sensors and uses a smart vision model to act like a partner rather than just a camera. It filters out the clutter to offer a helpful Detective Mode that highlights what actually matters based on what the user is trying to achieve right now.

Technology Stack and Implementation

Our method runs on a cycle where the system perceives, reasons, and acts. First, a fast layer spots basic objects, and then a deeper reasoning agent checks them against a knowledge graph to find strange behavior or risks. Finally, the system maps these findings directly into the view of the user. It places 3D markers on the display to show critical details like a safe exit or a broken part, keeping the user focused on the task without needing to search for answers manually.

Literature Review & Academic References (2024–2025)

- Key Research Papers:
    - “Disaster Copilot: A Vision for a Multi-Agent Artificial Intelligence System” (Camps-Valls et al., 2025) – Discusses the shift from passive models to active intelligent environments in disaster management.
    - “Context-Aware Perception: VLM-Augmented All-Weather Detection for Autonomous Driving” (Polin et al., 2025) – Explores using Vision-Language Models to handle visual degradation and complex scene reasoning.
    - “Feature4X: Bridging Any Monocular Video to 4D Agentic AI with Versatile Gaussian Feature Fields” (2025) – Research on creating "agentic" 4D representations of the physical world.

- Essential Books:
    - The Agentic AI Bible by Thomas R. Caldwell (2025): A foundational text on the principles of goal-driven autonomous agents.
    - Agentic Artificial Intelligence by Pascal Bornet and Jochen Wirtz (2025): Covers the impact of "Actionable AI" across industries.
    - Building Agentic AI Systems (Packt Publishing, 2025): A technical guide for developers on creating agents that can reason and plan.